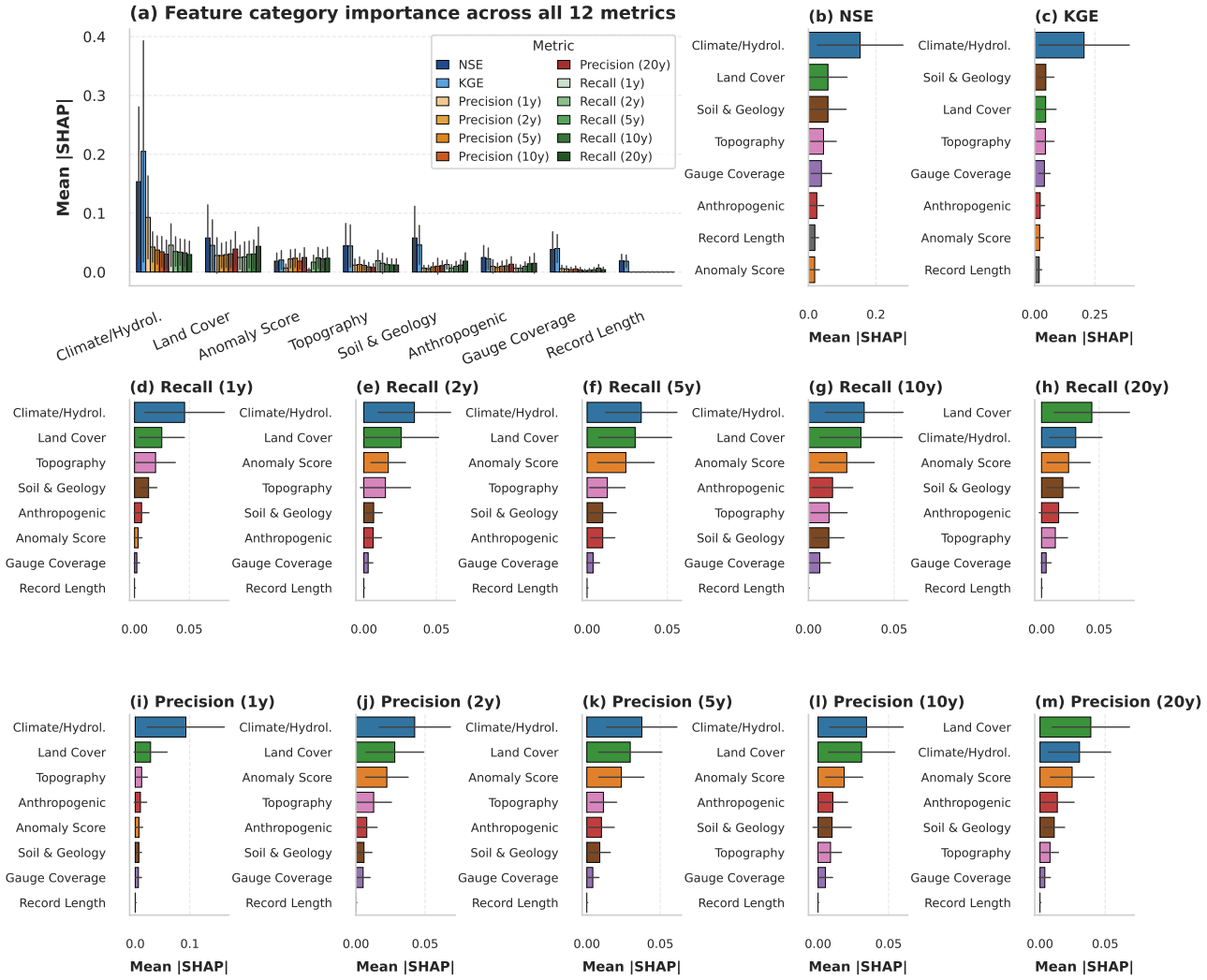
(a) Feature category importance across all 12 metrics
Mean |SHAP|
0.0
0.1
0.2
0.3
0.4
Metric
NSE
KGE
Precision (1y)
Precision (2y)
Precision (5y)
Precision (10y)
Precision (20y)
Recall (1y)
Recall (2y)
Recall (5y)
Recall (10y)
Recall (20y)
Climate/Hydrol.
Land Cover
Anomaly Score
Topography
Soil & Geology
Anthropogenic
Gauge Coverage
Record Length
(b) NSE
(c) KGE
0.0
0.2
0.00
0.25
Mean |SHAP|
(d) Recall (1y)
(e) Recall (2y)
(f) Recall (5y)
(g) Recall (10y)
(h) Recall (20y)
0.00
0.05
(i) Precision (1y)
(j) Precision (2y)
(k) Precision (5y)
(l) Precision (10y)
(m) Precision (20y)
0.0
0.1